Recall our ultimate goal in using BayesOpt: to search within the domain D of a black-box function the location at which the function is maximized. This means a natural search target for us is the location x^* that maximizes the objective value of the function f. That is, $f^* = f(x^*) \geq f(x)$, for all x in D.

IMPORTANT **The optimizer of a function**

The optimal location x* is often called the optimizer of the objective function f.

Given a Gaussian process (GP) belief about the objective function f, there is a corresponding probabilistic belief about the optimizer x^*.

Figure 6.10 shows an example of a trained GP and the distribution of the objective optimizer x^* that is induced from the GP. We note a few interesting characteristics of this distribution:

- The distribution is complicated and multimodal (having several local optima).
- The most likely location for the optimizer x^* is a bit to the left of zero. This is where the predictive mean of the GP is maximized.
- There is non-neligible probability that the optimizer x^* is at an end point -5 or 5.
- The probability that x^* is around 2 is almost zero, corresponding to the fact that we already have observed a higher objective value than $f(2)$.

These characteristics make modeling this distribution of x^* very challenging. The most simple method of gauging how this quantity x^* is distributed is to simply draw *many* samples from the Gaussian process and record the location at which each sample is maximized. This is in fact how Figure 6.10 is generated.

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